

The World's First Community-First Data Center

Carbon negative · Infrasound monitored · Closed-loop water · Waste-heat food production · Wildlife habitat · Community governed

Site 1: Kingsboro 900 MW

Site 2: Bennettsville 3 GW

World-First Infrasound Monitoring

Carbon Negative

1. ENVIRONMENTAL — Carbon · Sound · Wildlife · Air · Water

Sound, Vibration & Infrasound

World First: No data center on earth publicly monitors infrasound (1–20 Hz). We do it before the first turbine fires. Pre-construction baseline = legal fortress.

L1 – Acoustic Louvers	All turbine/HVAC inlets. –15 to –25 dB.
L2 – MLV Blankets	Mass-loaded vinyl on generators. STC 32–35.
L3 – MLV Perimeter Wall	15–20 ft fence, 4,000 LF. –20 to –30 dB.
L4 – Grass Berm Valley	Sloped 15 ft earth berm + native evergreens. –5 to –15 dB. Stormwater detention valley between berm and fence.
L5 – Rubberized Asphalt	All interior roads. Tire noise –6 to –10 dB.
L6 – Roof Acoustic Panels	MLV + mineral wool in all building roofs. –25 to –40 dB internal noise.
Infrasound Array	12 sensors: 4 on-site + 8 in community (schools, homes, park, church). All data streamed publicly. Pre-construction 6-month baseline.
Window Rebate	\$2,000 rebate for triple-pane upgrades within 0.5-mile radius. Budget: \$200K, covers ~100 homes.
Install Cost	\$196,000 Annual monitoring: \$28,000/yr

Carbon Negative Strategy

Goal	Not carbon neutral — carbon negative. More CO ₂ removed from atmosphere than the site generates, permanently.
Carbon-Seq Interiors	All desks, cabinets, wall panels, and flooring use carbon-sequestering materials: biochar-enhanced concrete, mass timber, hempcrete, mycelium composites. Each ton stores CO ₂ for 50–75 years.
Tree Planting Math	5,000 native trees on-site (~120 tons CO ₂ /yr). 50,000 trees via off-site reforestation partnership. ~42 trees offset 1 ton CO ₂ .
NC Native Species	Loblolly pine, Eastern red cedar, Tulip poplar, White oak, River birch, Sweetgum, Black walnut.

Carbon Credits	2,000–5,000 tons CO ₂ e/yr verified removal. At \$25–50/ton: \$50K–\$250K/yr additional revenue.
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Wildlife & Habitat Protection

Pre-Construction Survey	Full Phase 1 ecological survey: birds, pollinators, amphibians, mammals. All documented before ground breaks.
Wildlife Corridor	40-acre community green zone as continuous wildlife corridor. Unlit zones at night on eastern side for nocturnal movement.
Detention Pond Habitat	3 ponds engineered as wetland habitat: native emergent plants, wildlife shelves, no-fishing buffer zones.
Bird Strike Prevention	UV-pattern fritted glass on all building facades. Reduces bird-strike mortality by 50–90%.
Pollinator Strips	2-acre native wildflower and clover meadow along US-64 frontage. Monarch butterfly waystation certified. No pesticides — ever.
Light Pollution	Warm spectrum (2700K), downward-facing full-cutoff fixtures. Industrial zones: motion-activated only.

Air Quality & Pollution Control

Turbine Emissions	Selective catalytic reduction (SCR) on all turbine exhausts. Reduces NO _x by 80–90%. CEMS required by EPA.
WtE Emissions	Secondary combustion chamber at 950–1100°C destroys dioxins. Dry sorbent injection. Fabric filter for particulates.
Public Sensor Network	12 air quality sensors: PM _{2.5} , PM ₁₀ , NO _x , SO _x , CO, VOC, ozone. Live public dashboard. Third-party annual audit.
Living Air Filter	Green walls on turbine enclosure exteriors. Study-backed PM _{2.5} reduction 20–40% at wall surface.

Stormwater & Water Quality

Design Target	Zero net increase in stormwater runoff vs. pre-development. All rainfall managed on-site.
Berm Valley System	MLV berm set back 30 ft from fence, creating a grass valley that acts as a linear detention pond during rain events.
Bioswale Network	5-acre bioswale belt on US-64 frontage. Infiltrates 100% of light rain events. Native grasses + river cobble check dams.
Permeable Paving	All parking lots, walkways, and plaza: permeable interlocking concrete pavers (PICP). Recharges local aquifer.
Water Reuse	HVAC condensate + cooling tower blowdown collected into 500,000-gallon cistern. Reused for irrigation + toilet flushing. Reduces potable demand ~60%.

2. FOOD & WATER — Closed-Loop Aquaponics · Greenhouse Network

Closed-Loop Water System — Entire Site

90–95% less water vs. soil farming	~1% daily water loss (transpiration only)	500K gal on-site cistern from HVAC + rain	~Zero discharge to environment
Greenhouse Loop	NFT hydroponic channels recirculate nutrient solution. Aquaponics: fish tanks → biofilter → plant rafts → back to fish. True closed loop.		
Condensate Capture	Data center HVAC produces 5,000–15,000 gallons/day of condensate in NC humidity. Captured and routed to greenhouse top-off.		
Rainwater Harvesting	155-acre site at 46 in/yr = ~218M gal/yr potential. 10% capture = 21.8M gal/yr — far exceeds site demand.		

Greenhouse Network

GH1: Leafy Greens & Herbs 20,000 sq ft · NFT hydroponic rails · Lettuce, spinach, kale, basil, microgreens · ~150,000 lbs/yr	GH2: Fruiting Crops 15,000 sq ft · Dutch bucket drip · Tomatoes, cucumbers, peppers, strawberries · ~120,000 lbs/yr
GH3: Mushrooms 8,000 sq ft · Substrate growing rooms · Oyster, shiitake, lion's mane · ~40,000 lbs/yr	GH4: Root Crops + Towers 10,000 sq ft · DWC + vertical towers · Sweet potato, beets, radish, carrots · ~80,000 lbs/yr

Aquaponics + Marine Tank System

Fish & Shrimp RAS	20,000 sq ft recirculating aquaculture. Tilapia + Pacific white shrimp. ~25,000 lbs fish + 8,000 lbs shrimp/yr.
Marine Showcase	5,000-gallon display tank visible from walking path. Marine fish, corals, live rock, plankton cultures. Educational centerpiece.
Sushi-Grade Output	Zero antibiotics, clean water, zero-stress environment. Sushi-grade certification achievable. Weekly farm stand sales.
Plankton Cultivation	Controlled phytoplankton tanks feed shrimp larvae. Harvested as omega-3 supplement product.

~390K lbs annual produce yield	~33K lbs fish + shrimp per year	~875 people fully fed on produce alone
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3. ENERGY & WASTE — Solar · BESS · Waste-to-Energy

Solar Array Strategy

Rooftop Solar	All data center building roofs: bifacial monocrystalline panels. ~40 MW estimated from roof surfaces alone.
Carport Solar	Parking lot canopy solar — shade + power generation. ~8–12 MW additional. EV chargers integrated in posts.
Agrivoltaic Array	Panels elevated 8 ft on southern property line — pollinators and native grasses underneath. ~15 MW additional.
Total Solar	~60–70 MW peak (DC). NC averages 4.5 peak sun hours/day = ~100 GWh/yr. Powers 100% of community zone needs.
BESS	Grid-scale LiFePO4 battery storage. Sized to run community zone, greenhouses, and WtE for 8 hrs without grid.
Virtual Net Metering	Excess solar credited to local apartments and low-income households within 2-mile radius. 30–50% electricity bill reduction.

Waste-to-Energy System

The only self-funding program on the site. Tipping fees + energy + carbon credits generate \$2.7M/yr net after all operating costs.

Technology	MSW Gasification. Low-oxygen thermal decomposition → syngas → electricity. Lower NOx/SOx/particulates than incineration.
Capacity	100 tons/day (Phase 1). Franklin County generates ~44,000 tons/yr — site diverts ~80% of county waste from landfill.
Tipping Fee Revenue	\$45–75/ton (NC average). 100 tons/day x \$55 avg x 365 = ~\$2M/yr gate revenue alone.
Energy Output	2–3 MW continuous. ~\$1.1M/yr avoided energy cost.
Carbon Credits	Each ton diverted prevents ~0.5 tons methane CO ₂ e. ~\$456K/yr.

Build cost (Phase 1 gasification unit)	\$12.2M
Annual operating cost	\$999K/yr
Revenue: Tipping fees	+\$2.0M/yr
Revenue: Energy avoided/sold	+\$1.1M/yr
Revenue: Carbon credits	+\$456K/yr
NET ANNUAL REVENUE	+\$2.56M/yr

4. COMMUNITY — Plaza · Tours · Events

Community Event Plaza

Surface Design	3 acres permeable interlocking pavers. Flush power pedestals (120V/240V 30A) every 30 ft. Dual-purpose: daily parking + event venue.
Stage Infrastructure	Permanent 20x40 ft reinforced concrete stage pad with 400A electrical service, rigging anchor points, backstage utility room.
Acoustic Wing Walls	Two 12-ft permanent wing walls flanking stage — acoustic reflectors for events AND sound baffles facing US-64.
Restroom Building	Permanent 8-stall + 2 ADA + utility sink. Doubles as security office. No porta-potty rentals ever.
Farm Stand	Every Saturday 8am–1pm. Greenhouse produce, fish, shrimp at below-retail pricing. No registration. Open to all.
Site Tours	Free 45-min guided tours: infrasound demo, greenhouse walk-through, sound berm. Saturday 10am & 1pm, max 25/tour.
Build Cost	\$728,000 one-time \$46,000/yr operating

Annual Event Calendar

Weekly Farm Stand Every Saturday 8am. Fresh produce + fish + shrimp. Below-retail. Public. No registration.	Monthly Block Party 1st Saturday/month. Free. Live music, food trucks, kids activities. Priority to Vocational Academy grad food carts.
Quarterly Community Fair 4x per year. 1,500–2,000 attendees. Craft vendors, health screenings, career fair, live performances.	Annual SEED Day Once per year. Full campus open house. Tours, demos, community awards, graduate showcases.
School Science Tours Year-round Tue–Thu. K–12 groups. Infrasound demo, aquaponics lab, energy systems. Free.	SEED Night Market Summer Fridays. Evening market. Greenhouse produce, prepared foods, music. Sunset to 10pm.

5. VILLAGES — Dignity · Sanctuary · Re-Entry

Dignity Village — Homeless Re-Entry

100 tiny homes (200–400 sq ft)	\$5.7M build cost	\$3.8M annual operating cost	\$54K state savings per resident housed
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Residents	100 adults experiencing homelessness. 1 staff : 5 resident ratio.
Day Staff	10 case managers/vocational coaches, 4 security guards, 2 clinical support, 3 maintenance, 2 admin, 3 food services = 24 FTE
Night Staff	2 residential monitors, 4 security guards, 1 clinical on-call = 7 FTE overnight, always awake
Key Features	24/7 security, vocational training pipeline, community garden plots per resident, path to permanent housing
NC Context	11,626 homeless in NC (2024, +19% from 2023). State has 2,536-bed shelter deficit.

Sanctuary Village — Therapeutic Micro-Town

50 high-acuity MH residents	\$10M build cost (clinical spec)	\$4.2M annual operating cost	\$80K state savings per hospitalization prevented
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Residents	50 adults with serious mental illness (SMI). 1 staff : 3 resident ratio required.
Day Staff	8 LCSWs/therapists, 9 psychiatric techs, 1 psychiatrist (PT), 2 NPs, 4 security, 2 OTs, 3 recreation staff, 4 food services = 37 FTE
Night Staff	5 psychiatric technicians (awake), 4 security guards = 9 FTE overnight, always awake
Amenities	Grocery, diner, bowling alley, fitness cage, VR exergame studio, hydroponic garden therapy, park, sensory rest rooms
NC Context	356,000 NC adults live with serious mental illness. Only 45% receive any care.

Re-Entry City — Recidivism Reduction (Long-Term Vision)

200–1,000 beds (phased, start with 200)	\$18M 200-bed pilot build cost	\$6.5M annual operating (200 beds)	\$4.3M annual state savings (40% success rate)
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NC Need	18,000 released from NC prisons/yr. Recidivism rate 40–49%. \$54,000/yr to incarcerate = NC pays ~\$400M+/yr.
Day Staff	20 case managers, 8 vocational instructors, 6 substance counselors, 4 MH counselors, 8 security, 3 medical, 4 employment specialists = 67 FTE
Night Staff	6 residential monitors, 8 security guards = 14 FTE overnight, always awake
Funding	DOJ Second Chance Act grants, HHS Medicaid reentry waiver (2023), Social Impact Bonds, NC Reentry 2030 strategic funds.

6. CAPITAL SUMMARY — Full Cost Breakdown

Environmental & Site

MLV Perimeter Fencing + Berms (all layers)	\$1,400,000
Roof Acoustic Panels (all buildings)	\$380,000
Infrasound Monitoring Network (12 sensors)	\$196,000
Air Quality Sensor Network (12 sensors)	\$145,000
Window Rebate Fund (100 homes, 0.5 mi)	\$200,000
Stormwater: Bioswales, permeable paving, ponds	\$1,200,000
5,000 native trees on-site	\$375,000
50,000 trees off-site reforestation	\$500,000
Wildlife habitat design + bird-safe glass	\$180,000
Carbon-sequestering interior materials	\$850,000
Green walls on turbine enclosures	\$120,000

Food, Water & Aquaponics

4 Greenhouse structures	\$1,200,000
Hydroponic systems (NFT, DWC, Dutch bucket)	\$380,000
Aquaponics RAS system (fish + shrimp)	\$420,000
Marine showcase tank + plankton lab	\$85,000
Waste-heat exchange plumbing + distribution	\$280,000
LED grow lighting + climate automation	\$300,000
500,000-gal cistern + treatment system	\$180,000
Wash-pack + cold storage + farm stand	\$220,000

Energy & Waste

Rooftop solar (~40 MW, bifacial)	\$36,000,000
Carport solar (~10 MW)	\$8,000,000
Ground-mount agrivoltaic array (~15 MW)	\$9,000,000
BESS (8-hr community zone storage)	\$4,500,000
WtE Gasification Unit (100 tons/day)	\$8,000,000
WtE sorting facility + receiving building	\$1,800,000
WtE emissions control + odor system	\$580,000
Haul road + truck turnaround	\$340,000

Community Infrastructure

Event Plaza (permeable paving + power grid)	\$420,000
Stage pad + electrical + anchor points	\$85,000
Permanent restroom building (ADA)	\$145,000
Solar LED streetlights (150 poles)	\$2,100,000
ADA sidewalks + RRFB beacons (3 miles)	\$1,400,000

Smart traffic signals (8 intersections)	\$900,000
Public Wi-Fi mesh (150 nodes)	\$1,200,000

Villages (Pilot Scale)

Dignity Village (100 tiny homes + staff housing)	\$5,700,000
Sanctuary Village (50 residents, clinical spec)	\$10,000,000
Vocational Academy (classrooms + labs)	\$900,000
Security + perimeter tech	\$600,000
Recreation hub (gym, VR, bowling)	\$1,200,000

Annual Operating Summary

Village Operations (Dignity + Sanctuary + Vocational)	\$8,000,000
Greenhouse + Aquaponics Operations	\$500,000
WtE Operations	\$999,000
Community + Events + Plaza	\$150,000
Environmental Monitoring + Compliance	\$250,000
Security (all zones, 24/7)	\$1,200,000
Tree maintenance + landscaping	\$180,000
TOTAL ANNUAL OPEX	\$11,279,000
WtE Net Revenue	-\$2,560,000
Greenhouse Sales + Fish/Shrimp	-\$560,000
5% DC Revenue Endowment (est.)	-\$5,000,000+
NET COMMUNITY COST (after revenue)	~\$3.2M/yr

7. OFF-SITE COMMUNITY INVESTMENT — Beyond the Fence Line

Total program capital	~57% Grant / tax credit funded	~\$15.4M Net SEED investment	5,000+ Community members served
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Infrastructure Upgrades

Underground Utility Lines

Target Area	8–12 miles of residential feeder lines along US-64 corridor within 2-mile radius.
Cost	\$1.5M/mile x 10 miles = \$15M. FEMA HMGP grant (75%) = \$11.25M covered. SEED net: ~\$3.75M.
Funding	FEMA Hazard Mitigation Grant Program (75%), USDA ReConnect, Duke Energy partnership.

Smart Traffic Signals (20 intersections)

Technology	Adaptive Traffic Signal Control (ATSC): radar + video analytics. Real-time counting. Signal timing auto-adjusts every cycle.
Emergency Preemption	Fire truck or ambulance within 0.5 miles auto-clears every light on their route. Zero delay in emergencies.
Cost	\$120,000–\$180,000/intersection. 20 intersections = \$2.4–\$3.6M. Annual software/maintenance: \$80K/yr.

ADA Sidewalk Network — 40 Miles of Connectivity

Priority Routes	1. Housing to bus stop. 2. Housing to grocery store. 3. Housing to SEED campus. 4. School routes K–12.
Standard	5-ft minimum ADA compliant concrete. Detectable warning strips at all curb cuts. 30-year lifespan.
Phase 1 Cost	15 priority miles x \$200K = \$3,000,000
Full Build	40 miles x \$200K = \$8,000,000 (over 3–5 years)
Funding	USDOT Safe Streets & Roads for All (SS4A, up to \$25M), HUD CDBG, NC DOT TASA.

Street Trees + Solar LED Lighting

Street Trees	2,500 trees Phase 1 (1 per 30 ft of sidewalk). NC native species. ~\$1.5M. Property value +5–15% per USDA Forest Service.
Solar LED Lights	450 total poles (300 new + 150 replacement). Off-grid LiFePO4 battery. Zero utility bill forever. ~\$4.5M.
Funding	USDA Urban Forestry grants; EPA Solar for All, USDA REAP for lighting.

RO Water Treatment Plant — 0.5 MGD

99%+ PFAS removal	0.5 MGD capacity (500K gal/day)	~5,000 people served daily	\$110K net annual cost (after kiosk revenue)
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System Design	Pre-filtration → RO membrane array → UV disinfection → Remineralization → Distribution. All stages continuously monitored.
Community Kiosks	8 water dispensing kiosks in low-income housing areas. Clean water at \$0.01–\$0.02/gallon.
Power	~2,000 kWh/day powered entirely by on-site BESS + solar. Zero grid draw.

Total Capital	\$2,340,000 Funding: EPA WIFIA, USDA Rural Water grants, EPA PFAS remediation grants.
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Solar + Rent Stabilization Programs

Home Solar (300 homes)	\$0 upfront to homeowner. 30–50% electricity bill reduction. 5kW system + battery. ITC (30%) + USDA REAP (25%) fund ~55% of cost. Phase 1: ~\$5.25M, net ~\$1.6M after grants.
Apartment Solar (10 complexes)	Solar covers common area power. Remaining surplus split among tenants via VNM. Net SEED cost Phase 1: ~\$688,500.
3-Year Rent Cap Covenant	Property owners sign voluntary deed-recorded covenant: max 2.5%/yr rent increase. In exchange: solar install + community infrastructure upgrades raise property value 5–15%. NC legal (voluntary private covenant, not government rent control).
CLT Long-Term Play	SEED's Community Land Trust purchases key parcels within 0.5 miles. CLT land = permanent affordability removed from speculative market forever.

Off-Site Full Cost Summary

Underground Utility Lines (10 miles)	\$15,000,000 ~\$3.75M net
Smart Traffic Signals (20 intersections)	\$3,000,000 ~\$1.1M net
ADA Sidewalks (15 miles, Phase 1)	\$3,000,000 ~\$1.2M net
Street Trees (2,500 trees, Phase 1)	\$1,500,000 ~\$875K net
Solar LED Street Lighting (450 poles)	\$4,500,000 ~\$2.3M net
RO Water Treatment Plant (0.5 MGD)	\$2,340,000 ~\$1.5M net
Home Solar + Battery (300 homes)	\$5,250,000 ~\$2.1M net
Apartment Solar (10 complexes)	\$1,530,000 ~\$689K net
TOTAL OFF-SITE PORTFOLIO	~\$36.1M ~\$15.4M net